

World GIS Day 2025 Map Challenge Submission

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1. Introduction to the Problem

In Nigeria, one of the biggest threats to public health is not just disease, but safe access to care. When an area has high conflict, doctors cannot travel there, and vaccines cannot be delivered. Therefore, conflict is a physical barrier that blocks "pathways to health," leaving vulnerable communities cut off from life-saving medical aid.

2. Objective of the Map

The goal of this map is to visually identify "High-Risk Health Zones." By overlaying conflict intensity with health facility locations, we aim to spot exactly which hospitals and clinics are located in danger zones. This helps aid organizations know where they need security escorts to deliver medicine safely.

3. Data Sources Used

- **Conflict Data:** ACLED (Armed Conflict Location & Event Data Project) - Used to find where violence happened.
- **Health & Administrative Data:** eHealth Africa - Used for the map boundaries and the locations of health facilities.

4. Tools and Methods Applied

I used QGIS to build this solution:

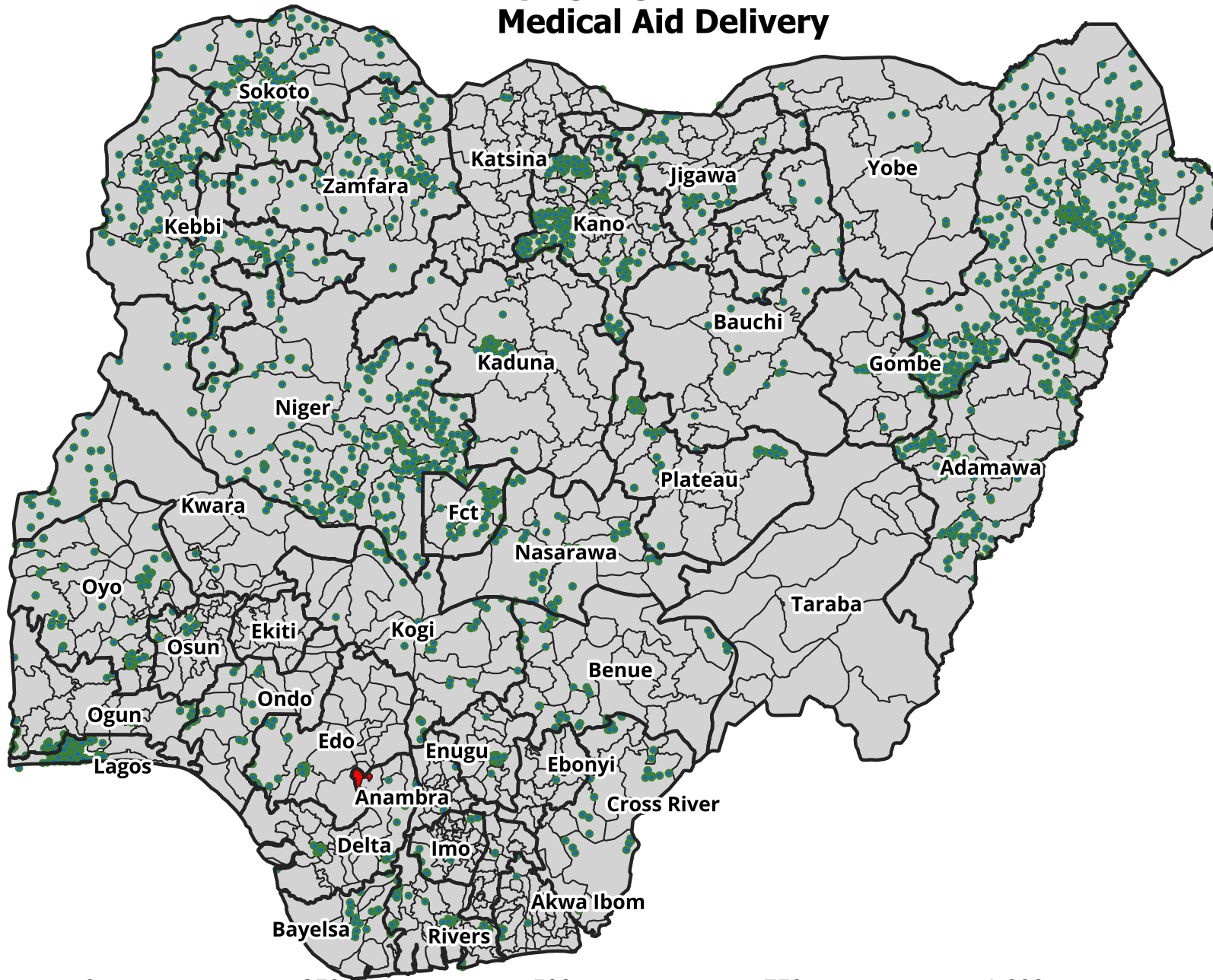
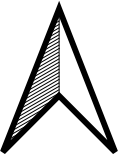
1. **Data Cleaning:** I filtered the raw conflict data to remove empty records.
2. **Analysis:** I used the "Statistics by Category" tool to count the total number of deaths for every Local Government Area (LGA).
3. **Visualization:** I used a "Graduated" color style (Red) to show conflict intensity and overlaid Health Facilities (Blue dots) to show the relationship between violence and care.

6. Insights Revealed

The visualization reveals a critical insight: While health facilities are spread across the country, those in the North-East (specifically Borno state) are trapped inside "Dark Red" conflict zones. This means hundreds of hospitals are operating in high-risk areas. Policy makers can use this map to prioritize these specific routes for military-escorted medical supply chains.

Barriers to Health Access: Conflict Intensity & Health Facilities in Nigeria

Identifying High-Risk Zones for Medical Aid Delivery



LEGEND

• Health Facilities

Number of Deaths

0 - 0

0 - 3

3 - 5

0 250 500 750 1,000 1,250 km

Source: eHealth Africa, ACLED

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